

Course Title: Physics I
Course Code: PHY 05331104
Credit: 3.0 Theory

Rationale:

This course aims to introduce the static and dynamics of particles, continuous oscillations with different amplitude scales as wind and rain induce in large structures (sky-scrapers, high towers and stayed cables in bridges). Engineers employ thermodynamics and optics knowledge in wastewater treatment, heating, ventilation, and air conditioning and fiber optics sensors.

Course Learning Outcomes:

At the end of this course, students will be able to:

CO1: Apply the knowledge of thermodynamics in ventilation and air conditioning equipment.

CO2: Apply Newton's law of motion to design engineering equipment's.

CO3: Understand the basic principle of different optical instruments.

CO4: Understand the difference between damped and forced harmonic oscillation.

Course Learning Outcomes (COs)	COURSE CONTENTS	Teaching Learning Strategy
Apply the knowledge of thermodynamics in ventilation and air conditioning equipment.	Heat and temperature; Kinetic theory of gases: mean free path, equipartition of energy, Brownian motion, Van der Waal's equation; zeroth law of thermodynamics, First Law of thermodynamics and its application, isothermal, isobaric and adiabatic process; reversible and irreversible processes. Second Law of thermodynamic, Efficiency of heat engines, Entropy and Disorder, third Law of Thermodynamics.	Lectures, Power point Presentation, Group discussion.
Apply Newton's law of motion to design engineering equipment's	Basic concepts of mechanics: Linear and circular motion; Application of Newton's Law of motion; centre of mass; Impulse and momentum; torque; centripetal force and centrifugal force; Friction; Moments of inertia of areas and masses; Principles of work-Energy theorem	Lectures, Power point Presentation, Group discussion.

Understand the basic principle of different optical instruments.	Wave front, Interference of light, Young's double slit experiment, Diffraction of light: Fresnel and Fraunhofer diffraction, Polarization.	Lectures, Power point Presentation, Group discussion.
Understand the difference between damped and forced harmonic oscillation.	Wave, Differential equation of a simple harmonic oscillator, total energy and average energy, Simple pendulum, damped oscillation, forced oscillation, progressive wave, resonance, stationary wave, group velocity and phase velocity.	Lectures, Power point Presentation, Group discussion.

Recommended Books:

1. Resnick, Robert. & Halliday, David, Physics, New York: Wiley; 1966.
2. Beiser, A. Concepts of Modern Physics, 2nd ed., New York: McGraw-Hill; 1973.
3. Dr. Gias Uddin Ahmed, Physics for engineers (Part-1), 1st ed., Hafiz Book Center; 2005.

CO and PO relationship:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	Bloom's Taxonomy
CO1	√	√											C1
CO2	√	√											C2, C3
CO3	√				√								C3, C4, C5
CO4	√	√											C4, C5
BNQF Skills											F1, F2, T2, T4, L1, L2		

Program Outcomes:

PO1: Engineering Knowledge	PO5: Modern Tools	PO9: Communication
PO2: Problem Analysis	PO6: Engineer and Society	PO10: Team Work
PO3: Design / Development	PO7: Sustainability	PO11: Project Management
PO4: Investigation	PO8: Ethics	PO12: Lifelong Learning

Bloom's Taxonomy:

Cognitive Domain (Knowledge based)	Affective Domain (Emotive based)	Psychomotor Domain (Action based)
C1: Remembering	A1: Receiving	P1: Perception
C2: Understanding	A2: Responding	P2: Set
C3: Applying	A3: Valuing	P3: Guided Response
C4: Analysing	A4: Organizing	P4: Mechanism
C5: Evaluating	A5: Characterizing	P5: Complex overt response
C6: Synthesizing (Creating)		P6: Adaptation

BNQF Skills:

Fundamental Skills	Social Skills	Thinking Skills	Personal Skills
F1: Knowledge	S1: Communication	T1: Decision Making	L1: Organizing
F2: Apply	S2: Language Proficiency	T2: Technical Knowledge	L2: Ethics
F3: Critical Evaluation	S3: Entrepreneurship	T3: Managerial Knowledge	L3: Lifelong Learning
F4: Leadership	S4: Civic Literacy	T4: Team Work	L4: Society Contribution
F5: Digital Literacy			

Credit hours estimation:

Total Student Learning Time (SLT)	Face to Face (hours)	Total Guided and Independent Learning (hours)
Lecture	30 (2.5 hour x 12 weeks)	36 (3 hours x 12 weeks)
Tutorial	05 (0.5 hour x 10 weeks)	05 (0.5 hour x 10 weeks)
Course Assignment / Project/Forum/Case Study	02 (1 hour x 2 Assignment/Forum)	04 (2 hours x 2 Assignment/Forum)
Quizzes	01 (2 Quizzes x 0.5 hour)	04 (2 Quizzes x 2 hours)
Midterm	01 (1 Exam x 1 hour)	04 (1 Exam x 4 hours)

Term Paper/Final	02 (1 Final exam x 2 hours)	08 (1 Final exam x 08 hours)
Subtotal (THEORY)	41 Hours	61 Hours
Total (THEORY)	102 Hours (For 3.0 Credits)	

Complex Activities and Knowledge Profile:

COs	Complex Engineering Problem	Complex Activities	Knowledge Profile	Assessment
CO1	-	-	K6	Quiz, Assignment, Tutorial, Examination
CO2	-	-	K1, K2, K3, K4	Quiz, Assignment, Tutorial, Examination
CO3	-	-	K5	Quiz, Assignment, Tutorial, Examination
CO4	-	-	K5	Quiz, Assignment, Tutorial, Examination

Knowledge Profile:

	Attribute
K1	A systematic, theory-based understanding of the natural sciences applicable to the discipline.
K2	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modelling applicable to the discipline.
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
K5	Knowledge that supports engineering design in a practice area.
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
K7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability.
K8	Engagement with selected knowledge in the research literature of the discipline.

Range of Complex Engineering Problem Solving:

Attribute	Complex Engineering Problems have characteristic P1 and some or all of P2 to P7:
Depth of knowledge required	P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach.

Range of conflicting requirements	P2: Involve wide-ranging or conflicting technical, engineering and other issues
Depth of analysis required	P3: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.
Familiarity of issues	P4: Involve infrequently encountered issues.
Extent of applicable codes	P5: Are outside problems encompassed by standards and codes of practice for professional engineering.
Extent of stakeholder involvement and conflicting requirements	P6: Involve diverse groups of stakeholders with widely varying needs.
Interdependence	P7: Are high level problems including many component parts or sub-problems

Range of Complex Engineering Activities:

Attribute	Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:
Range of resources	A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies).
Level of interaction	A2: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues.
Innovation	A3: Involve creative use of engineering principles and research-based knowledge in novel ways.
Consequences for society and the environment	A4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches.

Course Outline

Week	Unit Learning Outcomes	COURSE CONTENTS
1.	Program Outcome (PO), Vision, Mission, Objectives. CO – PO relationship	Course Introduction
2.	Understand basic concept of Thermodynamics, Understand kinetic theory of gas	Heat and temperature, Kinetic theory of gases: mean free path, Gas equations,
3.	Explain Equipartition theorem of energy, Understand basic laws of thermodynamics	Equipartition of energy: Brownian motion, Van der Waal's equation, Zeroth law of thermodynamics, First Law of thermodynamics
4.	Understand efficiency of a heat engine, Apply Carnot's theorem	Reversible and irreversible processes. Second Law of thermodynamic, Efficiency of heat engines, Entropy and Disorder, third Law of Thermodynamics.
5.	Define different types of motion Understand displacement, velocity and acceleration	Basic concepts of mechanics: Linear and circular motion
MIDTERM		
7.	Apply Newton's laws of motion Apply linear momentum Apply angular momentum	Application of Newton's Law of motion: Centre of mass; Impulse and linear momentum; torque and angular momentum, centripetal force and centrifugal force
8.	Apply moment of inertia Understand work-energy	Friction: Moments of inertia of areas and masses; Work-Energy Principles of work-Energy theorem
9.	Understand different properties of light	Basic discussion on theory of light, Interference, Young's double slit experiment
10.	Apply diffraction and interference of light, Understand and apply polarization of light	Diffraction of light: Fresnel and Fraunhofer diffraction, Polarization of Light Brewster's law
11.	Understand different types of waves, Define damped, undamped and forced oscillation	Basic concepts of wave and oscillation , Simple pendulum, Loaded Spring, damped, Undamped and Forced Oscillation
12.	Simple harmonic motion and apply second order differential equation of motion of SHM	Simple Harmonic Motion, Second order differential equation of SHM and its application.
FINAL EXAMINATION		

Assessment & Evaluation:

Methods	Percentage (%)	COs
Attendance & Performance	10	1,2,3,4
Project/ Case Study/ Assignment/ Presentation	10	1,2,3,4
Forum/ Discussion	10	1,2,3,4
Quizzes	15	1,2,3,4
Midterm	15	1,2,3,4
Final Examination	40	1,2,3,4